Chapter 6: Maximum Likelihood Estimation

Part I — Statistics Foundation for ML

Omid Safarzadeh

Abstract

Maximum likelihood is the estimation principle underneath most of modern machine learning. Faced with data and a model that has an unknown parameter, it asks a single question — which value of the parameter would have made the data we actually observed most probable? — and takes the answer as the estimate. This chapter develops that idea in full: we define the likelihood, compute the estimator for the models you meet most often, and then ask how good it is, arriving at the score, the Fisher information, the Cramér–Rao lower bound, and the fact that the estimator is asymptotically normal. A final section looks at what happens when the usual assumptions fail, and shows that maximising the likelihood is the same as minimising a Kullback–Leibler divergence — the sentence that ties this chapter to every cross-entropy loss you will later minimise.

Contents

1 The likelihood and the maximum-likelihood estimator 1

2 Score, information, and the quality of the estimator 3

3 When the assumptions fail 5

1 The likelihood and the maximum-likelihood estimator

Suppose we observe data $x_1, \dots, x_n$, modelled as independent draws from a density (or mass function) $f(x\,;\theta)$ with an unknown parameter θ. The density answers a forward question: with θ fixed, how probable is each possible value of x? Estimation runs that question backwards. The data are now in hand and fixed; it is θ we do not know. Reading the same formula with the data fixed and θ free gives the **likelihood**: how probable were the numbers we actually saw, as a function of the parameter that might have produced them?

Definition 1.1 (Likelihood and log-likelihood)**.** For independent observations the *likelihood* is

$$L(\theta) = \prod_{i=1}^{n} f(x_i\,;\theta), \qquad \ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f(x_i\,;\theta).$$

The *maximum-likelihood estimator* (MLE) is the value that makes the observed data most probable,

$$\hat{\theta} = \arg\max_{\theta} L(\theta) = \arg\max_{\theta} \ell(\theta).$$

We almost always maximise the log-likelihood ℓ rather than L itself, and for three reasons. The logarithm is strictly increasing, so it does not move the location of the maximum; it turns the product into a sum, which we can differentiate term by term; and it replaces the multiplication of thousands of small probabilities, which underflows to zero in floating point, by the addition

of their logarithms, which does not. In the smooth case we then find the maximum the usual way: solve the **likelihood equation** $\ell'(\theta) = 0$ and confirm the solution is a maximum rather than a minimum or a saddle. The three examples below are the ones worth knowing by heart.

Example 1.1 (Bernoulli)**.** For $x_i \in \{0, 1\}$ independent and Bernoulli(p),

$$\ell(p) = \Big(\textstyle\sum_i x_i\Big) \log p + \Big(n - \textstyle\sum_i x_i\Big) \log(1-p), \qquad \ell'(p) = \frac{\sum_i x_i}{p} - \frac{n - \sum_i x_i}{1-p}.$$

Setting $\ell'(p) = 0$ gives $\hat{p} = \bar{x}$: the maximum-likelihood estimate of a probability is just the observed proportion.

Example 1.2 (Normal)**.** For x_i independent and $N(\mu, \sigma^2)$,

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2,$$

and solving the two likelihood equations (one for μ, one for σ^2) gives

$$\hat{\mu} = \bar{x}, \qquad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

Example 1.3 (Poisson)**.** For x_i independent and Poisson(λ) we have $\ell(\lambda) = \big(\sum_i x_i\big) \log \lambda - n\lambda - \sum_i \log(x_i!)$, so once again $\hat{\lambda} = \bar{x}$.

Every one of these estimators is a one-liner once you have the formula. Here are four of them side by side, in plain Python — the kind of side-by-side that makes maximum likelihood click:

```
def mle_bernoulli(x):    return sum(x) / len(x)          # p_hat      = mean
def mle_poisson(x):      return sum(x) / len(x)          # lambda_hat = mean
def mle_exponential(x): return len(x) / sum(x)           # rate       = 1 / mean
def mle_normal(x):
    mu  = sum(x) / len(x)
    var = sum((xi - mu)**2 for xi in x) / len(x)         # divide by n: the MLE
    return mu, var

print(mle_bernoulli([1, 1, 0, 1, 0]))          # -> 0.6
print(mle_poisson([2, 1, 3, 0, 2, 1, 4, 2]))   # -> 1.875
print(mle_exponential([0.4, 1.2, 0.9, 2.1]))   # -> 0.8695...
print(mle_normal([2, 4, 4, 4, 5, 5, 7, 9]))    # -> (5.0, 4.0)
```

The recipe never changes: write down $\ell(\theta)$, set $\ell'(\theta) = 0$, read off the estimator. The R versions sit beside these in the chapter's `exercises/` folder.

Editorial note

Notice that the normal variance estimate divides by n, not by $n - 1$. The maximum-likelihood estimator is therefore *biased*: $E[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2$, a little too small on average. This is not a mistake to be corrected — it is what maximum likelihood does. The method optimises the fit to the sample actually observed and makes no promise of unbiasedness; the next section makes the trade-off precise. (The deliberately planted bug in this chapter's first exercise is exactly this n-versus-$(n-1)$ slip.)

A cookbook of estimators

The same recipe — write down $\ell(\theta)$, solve $\ell'(\theta) = 0$ — produces a one-line estimator for most of the distributions you will meet. The common closed forms are collected below; a runnable version of every one, in Python and R, sits beside this chapter in `mle_cookbook.py` and `mle_cookbook.R`.

Distribution	Maximum-likelihood estimate
Bernoulli(p)	$\hat{p} = \bar{x}$
Binomial(k, p), k known	$\hat{p} = \bar{x}/k$
Geometric(p)	$\hat{p} = 1/\bar{x}$
Poisson(λ)	$\hat{\lambda} = \bar{x}$
Exponential(λ)	$\hat{\lambda} = 1/\bar{x}$
Normal(μ, σ^2)	$\hat{\mu} = \bar{x}, \quad \hat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \bar{x})^2$
Uniform($0, \theta$)	$\hat{\theta} = \max_i x_i$
Laplace(μ, b)	$\hat{\mu} = \text{median}(x), \quad \hat{b} = \frac{1}{n}\sum_i \lvert x_i - \hat{\mu}\rvert$
Pareto(x_m, α), x_m known	$\hat{\alpha} = n \big/ \sum_i \log(x_i/x_m)$
Gamma(k, θ), k known	$\hat{\theta} = \bar{x}/k$

When no closed form exists — the Gamma with both parameters free, the Weibull, the Beta — the recipe is unchanged; you solve $\ell'(\theta) = 0$ numerically, with Newton's method on the score. The cookbook includes one such example.

2 Score, information, and the quality of the estimator

Computing the estimator is only half the job. We also want to know how much to trust it: how sharply the likelihood peaks around $\hat{\theta}$, and whether any other estimator could do better. Both questions are answered by two quantities built from the derivative of the log-likelihood — the score and the Fisher information.

Definition 2.1 (Score and Fisher information)**.** The **score** is the derivative of the log-likelihood, $U(\theta) = \dfrac{\partial \ell}{\partial \theta}$. The **Fisher information** is its variance,

$$I_n(\theta) = \operatorname{Var}_\theta\big(U(\theta)\big) = E_\theta\big[U(\theta)^2\big].$$

For n independent observations $I_n(\theta) = n\, I_1(\theta)$, where I_1 is the information carried by a single observation.

The score measures how strongly the data push the estimate up or down at a given θ; at the maximum the push vanishes, which is just the likelihood equation $U(\hat{\theta}) = 0$. Two facts make the score the right object to study. The first is that, averaged over datasets drawn from the model at the true θ, it is zero.

Theorem 2.1 (The score has mean zero)**.** Under the regularity conditions of Section 3, $E_\theta[U(\theta)] = 0$.

Proof. Work with a single observation; summing over i gives the general case. Because $f(\cdot\,;\theta)$ is a density it integrates to 1 for every θ, so $\frac{\partial}{\partial\theta}\int f(x;\theta)\,dx = 0$. Exchanging the derivative and the integral — this is the regularity assumption — and using $\partial_\theta f = f\,\partial_\theta \log f$,

$$0 = \int \frac{\partial}{\partial\theta} f(x;\theta)\,dx = \int \big(\partial_\theta \log f(x;\theta)\big) f(x;\theta)\,dx = E_\theta\big[\partial_\theta \log f(X;\theta)\big] = E_\theta[U]. \qquad \square$$

Differentiating that identity once more gives a second fact, which is the reason the information can be computed two different ways — as the variance of the score, or as the curvature of the log-likelihood.

Theorem 2.2 (Information identity)**.** Under regularity, $I_1(\theta) = -\,E_\theta\big[\partial_\theta^2 \log f(X;\theta)\big]$.

Proof. Differentiate $\int (\partial_\theta \log f)\, f\, dx = 0$ with respect to θ:

$$0 = \int \big(\partial_\theta^2 \log f\big) f\, dx + \int \big(\partial_\theta \log f\big)\big(\partial_\theta f\big)\, dx = E\big[\partial_\theta^2 \log f\big] + E\big[(\partial_\theta \log f)^2\big].$$

The last term is $E[U^2] = I_1$, so $I_1 = -E[\partial_\theta^2 \log f]$: more curvature at the peak means more information. □

Information matters because it sets a hard floor on how well *any* unbiased estimator can do. No cleverness beats it.

Theorem 2.3 (Cramér–Rao lower bound)**.** Let $T = T(X_1, \dots, X_n)$ be an unbiased estimator of θ. Under regularity,

$$\operatorname{Var}_\theta(T) \;\geq\; \frac{1}{I_n(\theta)}.$$

Proof. Let $U = \sum_i \partial_\theta \log f(X_i;\theta)$ be the total score, so $E[U] = 0$ and $\operatorname{Var}(U) = I_n$. Unbiasedness means $E_\theta[T] = \theta$ for every θ; differentiating both sides and moving the derivative inside the integral,

$$\begin{aligned} 1 = \frac{d}{d\theta} E_\theta[T] &= \int T\, \partial_\theta\Big(\prod_i f\Big)\, dx \\ &= \int T\, U \Big(\prod_i f\Big)\, dx = E[TU] = \operatorname{Cov}(T, U), \end{aligned}$$

the last equality because $E[U] = 0$. The Cauchy–Schwarz inequality then gives

$$\operatorname{Cov}(T,U)^2 \leq \operatorname{Var}(T)\, \operatorname{Var}(U),$$

that is $1 \leq \operatorname{Var}(T)\, I_n$, which rearranges to the claim. □

Finally, with enough data the estimator settles into a normal distribution whose spread is exactly the Cramér–Rao floor — so the MLE is, in large samples, as good as an unbiased estimator can be.

Theorem 2.4 (Asymptotic normality of the MLE)**.** Under regularity, as $n \to \infty$,

$$\sqrt{n}\,(\hat\theta - \theta) \;\xrightarrow{\;d\;}\; N\Big(0,\; \tfrac{1}{I_1(\theta)}\Big).$$

Sketch. The estimator solves $U(\hat\theta) = 0$. Expanding the total score to first order about the true θ gives $0 = U(\theta) + U'(\theta^*)(\hat\theta - \theta)$ for some θ^* between θ and $\hat\theta$, hence

$$\sqrt{n}(\hat\theta - \theta) = \frac{\frac{1}{\sqrt{n}}\, U(\theta)}{-\frac{1}{n}\, U'(\theta^*)}.$$

The numerator is $n^{-1/2} \sum_i \partial_\theta \log f(X_i;\theta)$, a standardised sum of independent mean-zero terms each with variance I_1, so the central limit theorem sends it to $N(0, I_1)$. The denominator is an average that the law of large numbers sends to $-E[\partial_\theta^2 \log f] = I_1$ (and $\theta^* \to \theta$). Slutsky's theorem combines the two into a limit $N(0, I_1)/I_1 = N(0, 1/I_1)$. □

Example 2.1 (The Bernoulli MLE is efficient)**.** A single Bernoulli(p) observation has $I_1(p) = \frac{1}{p(1-p)}$, so $I_n(p) = \frac{n}{p(1-p)}$ and the Cramér–Rao bound is $\frac{p(1-p)}{n}$. The estimator $\hat{p} = \bar{x}$ has variance exactly $\frac{p(1-p)}{n}$: it sits on the floor for every n, not just asymptotically. An estimator that attains the Cramér–Rao bound is called **efficient**.

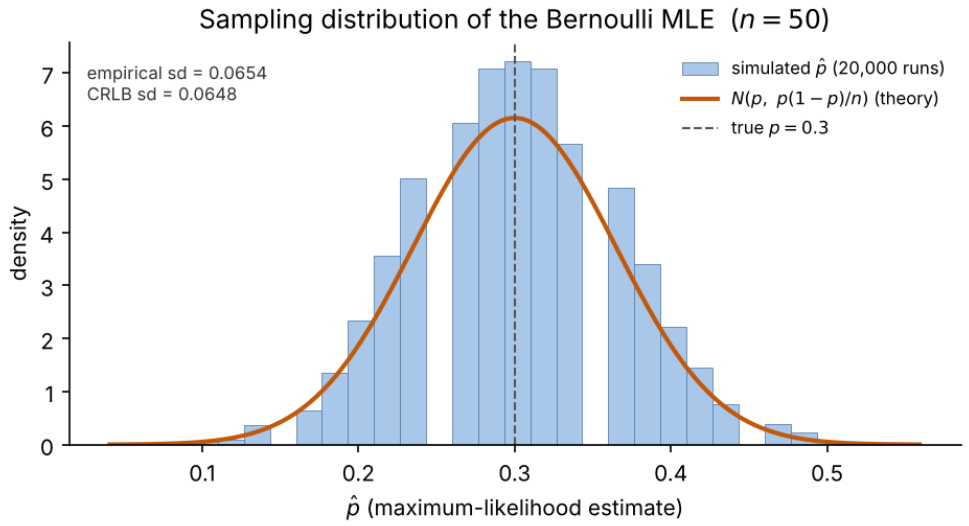
Asymptotic normality is easy to state and easy to doubt. It is also easy to just *watch* happen. The code below draws twenty thousand samples of size $n = 50$ from Bernoulli(0.3), takes the MLE of each, and compares the spread of those estimates against the Cramér–Rao floor:

```
import random

def bernoulli_mle(x): return sum(x) / len(x)

p, n, reps = 0.3, 50, 20000
rng = random.Random(0)
est = [bernoulli_mle([1 if rng.random() < p else 0 for _ in range(n)])
       for _ in range(reps)]

mean = sum(est) / reps
var  = sum((e - mean)**2 for e in est) / reps
print(var, p * (1 - p) / n)      # -> 0.004275   vs   CRLB 0.004200
```



The two numbers agree, and the picture says why: the histogram of the estimates is already indistinguishable from the normal curve the theorem predicts, and its spread sits on the Cramér–Rao floor. At $n = 50$ the asymptotic result is not a distant promise — it has already arrived.

3 When the assumptions fail

Every result above was stated "under regularity." It is worth seeing exactly what that phrase stands for, because it is not a formality: when it breaks, the conclusions break with it. The conditions are that the support of $f(\cdot\,;\theta)$ does not depend on θ; that $\log f$ is twice differentiable in θ; that the derivative and the integral may be exchanged; that $0 < I_1(\theta) < \infty$; and that the true θ lies in the interior of the parameter space.

Example 3.1 (Where regularity fails: Uniform$(0,\theta)$)**.** Here $f(x;\theta) = \frac{1}{\theta}\mathbf{1}\{0 \le x \le \theta\}$, and the support *does* depend on θ, so the step that exchanges the derivative and the integral is illegal. The MLE is $\hat{\theta} = \max_i x_i$, which is biased, converges at rate n rather than $\sqrt{n}$, and has an exponential rather than a normal limiting distribution. The Cramér–Rao bound does not apply at all. Regularity is doing real work in the theorems above.

There is also a broader way to read maximum likelihood that survives even when the model is wrong. Writing $m(x;\theta) = \log f(x;\theta)$, the MLE maximises the sample average $\frac{1}{n}\sum_i m(x_i;\theta)$. Estimators of this "optimise an average over the data" form are called **M-estimators**, and the MLE is the special case $m = \log f$. By the law of large numbers that average converges to $E_g[\log f(X;\theta)]$, where g is the *true* data-generating density — which need not be any member of our model. Maximising it is the same as minimising

$$D_{\mathrm{KL}}(g \,\|\, f_\theta) = E_g\Big[\log \tfrac{g(X)}{f(X;\theta)}\Big],$$

since the $E_g[\log g]$ piece does not involve θ. So the MLE aims at the parameter θ^* whose model is Kullback–Leibler closest to the truth. If the model is correct, θ^* is the true parameter; if it is wrong, the estimator still converges — to the best approximation the model can offer.

When the model is wrong the estimator is still asymptotically normal, but its spread is no longer the simple $1/I_1$.

Theorem 3.1 (Sandwich variance under misspecification)**.** When $g \notin \{f_\theta\}$,

$$\sqrt{n}(\hat{\theta} - \theta^*) \xrightarrow{d} N\big(0,\; A^{-1}BA^{-1}\big),$$

with $A = -E_g[\partial_\theta^2 m]$ and $B = \mathrm{Var}_g(\partial_\theta m)$. When the model is correct the information identity gives $A = B = I_1$ and the two factors collapse to the familiar $1/I_1$.

Summary

The MLE maximises $\ell(\theta) = \sum_i \log f(x_i;\theta)$. Its score has mean zero, its precision is measured by the Fisher information, no unbiased estimator can beat $1/I_n$ (Cramér–Rao), and in large samples it is $N(\theta, 1/I_n)$ — provided the regularity conditions hold and the model is correct. Read as an M-estimator minimising a Kullback–Leibler divergence, maximum likelihood is exactly what a classifier trained on cross-entropy, or a density model trained on negative log-likelihood, is doing. That is why this chapter sits under the loss-function chapter in Part II.